

Data Link Control

Topic

- Access Control
- Flow Control
- Error Control



Learning Outcome

- Explain the importance of protocol and data link control (A3)
- How how the access control being done(P2)
- Show how the flow control being done(P2)

Introduction

- The most important functions in the data link layer are:
 - Access Control – determines which who can send and when
 - Flow control – coordinates the amount of data that can be sent
 - Error control – perform error detection and correction



MAC Protocols: A taxonomy

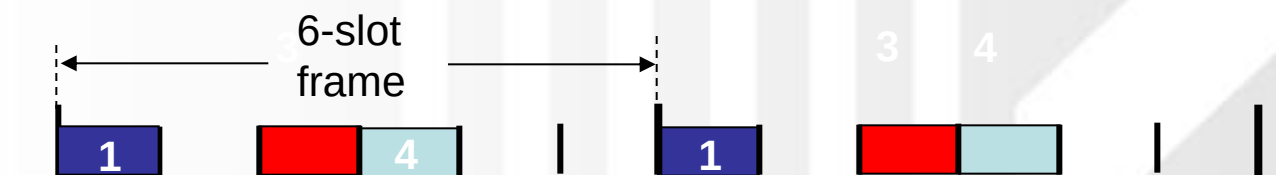
Three broad classes:

- **Channel Partitioning**
 - divide channel into smaller “pieces” (time slots, frequency) using FDM, WDM or TDM
- **Random Access**
 - channel not divided, allow collisions
 - “recover” from collisions
- **“Taking turns”/Polling**
 - nodes take turns, but nodes with more to send can take longer turns

Channel Partitioning

TDMA: time division multiple access

- access to channel in "rounds"
- each station gets fixed length slot (length = pkt trans time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle



Random Access Protocols

- When node has packet to send
 - transmit at full channel data rate R .
 - no *a priori* coordination among nodes
- two or more transmitting nodes → “collision”
- **random access MAC protocol** specifies:
 - how to detect collisions
 - how to recover from collisions
 - Examples protocols: slotted ALOHA, ALOHA, CSMA, CSMA/CD, CSMA/CA

ALOHA

- Developed for packet radio nets
- When station has frame, it sends
- Then listens for a bit over max round trip time
 - if receive ACK then fine
 - if not, retransmit
 - if no ACK after repeated transmissions, give up
- frame may be damaged by noise or collision

Figure 12.3 *Frames in a pure ALOHA network*

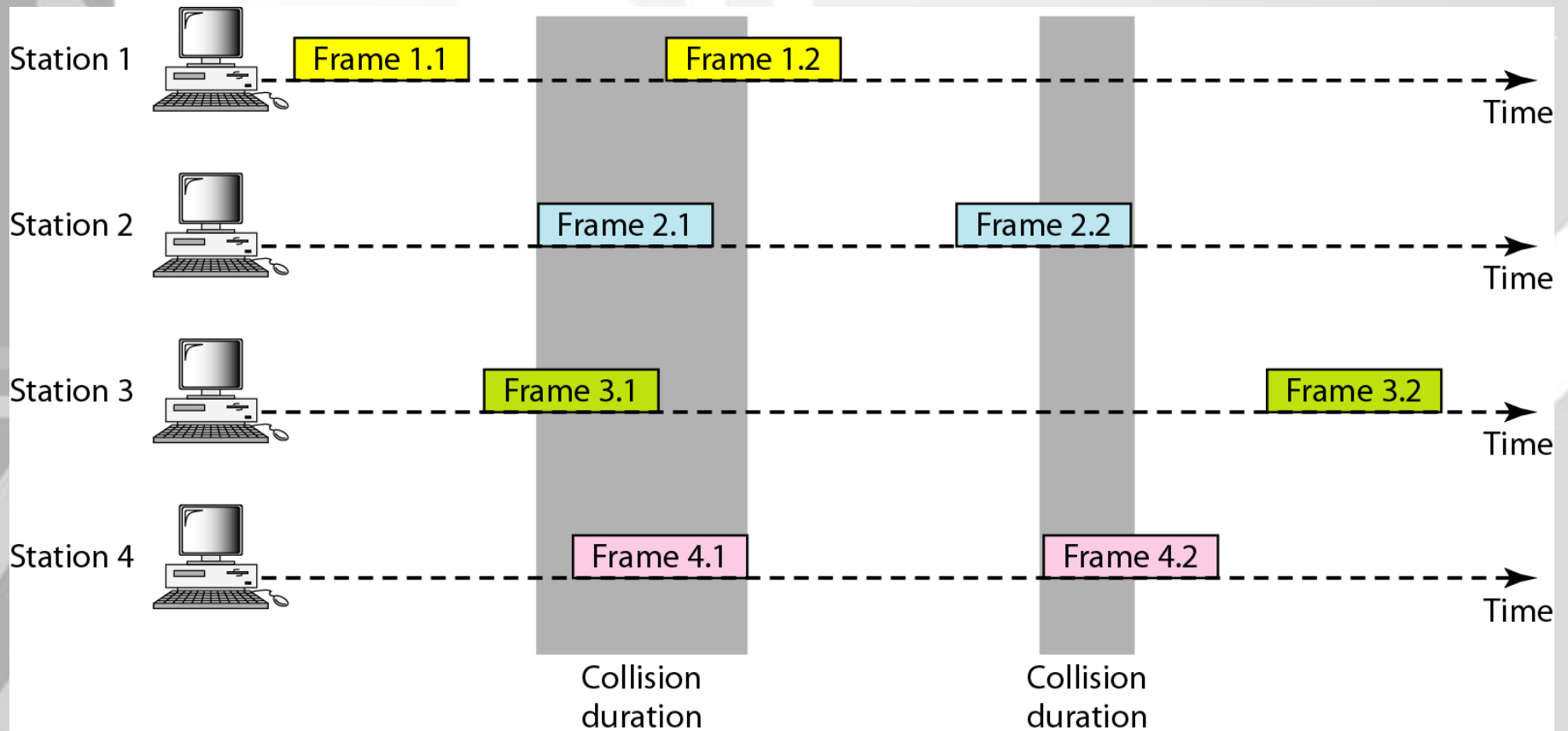
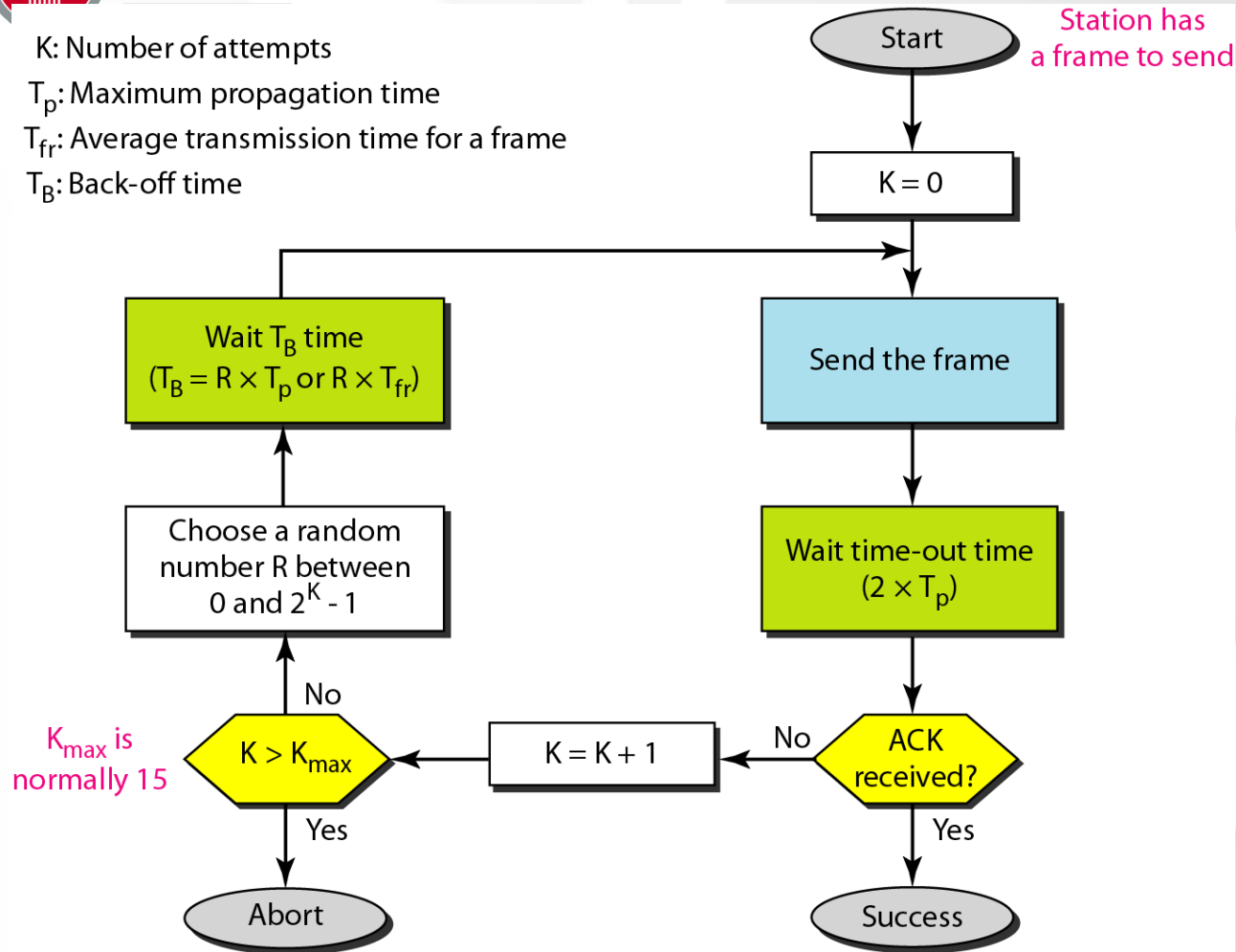


Figure 12.4 Procedure for pure ALOHA protocol

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



Slotted ALOHA

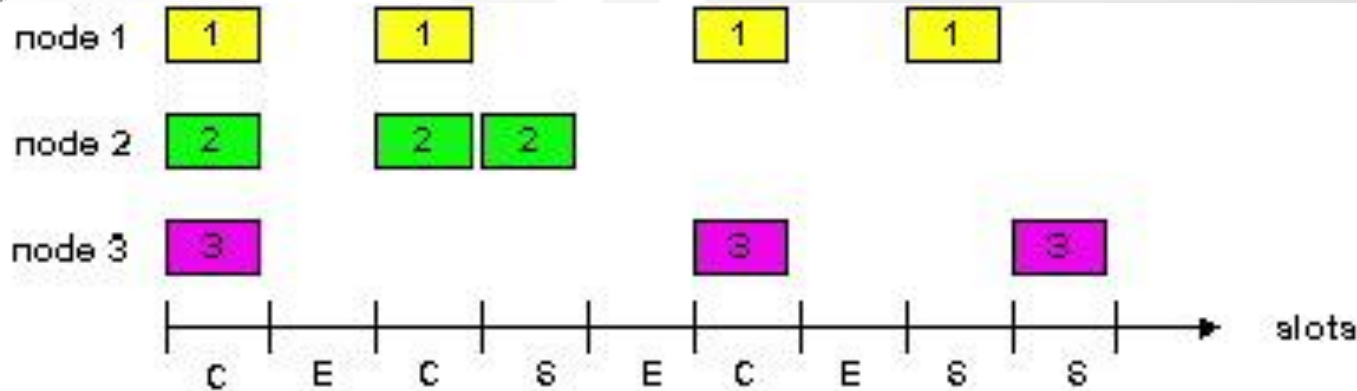
Assumptions:

- all frames same size
- time divided into equal size slots (time to transmit 1 frame)
- nodes start to transmit only slot beginning
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

Operation:

- when node obtains fresh frame, transmits in next slot
 - *if no collision*: node can send new frame in next slot
 - *if collision*: node retransmits frame in each subsequent slot with probability p

Slotted ALOHA



Pros

- single active node can continuously transmit at full rate of channel
- highly decentralized: nodes decide independently
- simple

Cons

- collisions, wasting slots
- idle slots
- clock synchronization

CSMA (Carrier Sense Multiple Access)

- stations soon know transmission has started
- so first listen for clear medium (carrier sense)
- if medium idle, transmit
- if two stations start at the same instant, collision
 - wait reasonable time
 - if no ACK then retransmit

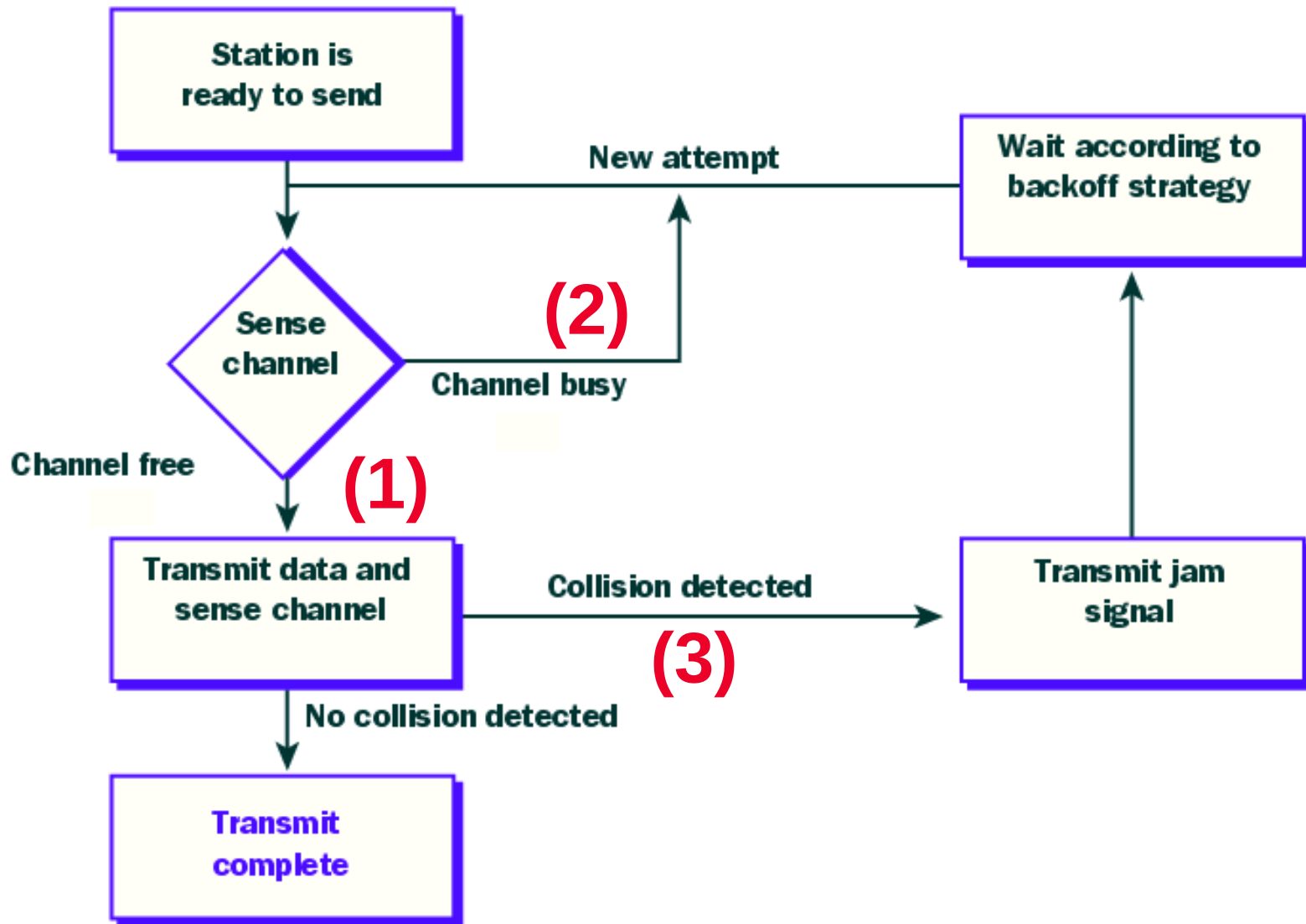
CSMA/CD Description

CSMA/CD rules:

1. if medium idle, transmit
2. if busy, listen for idle, then transmit
3. if collision detected, transmit jam signal and then stop transmission(**backoff**), wait random time then retry

Use by Ethernet

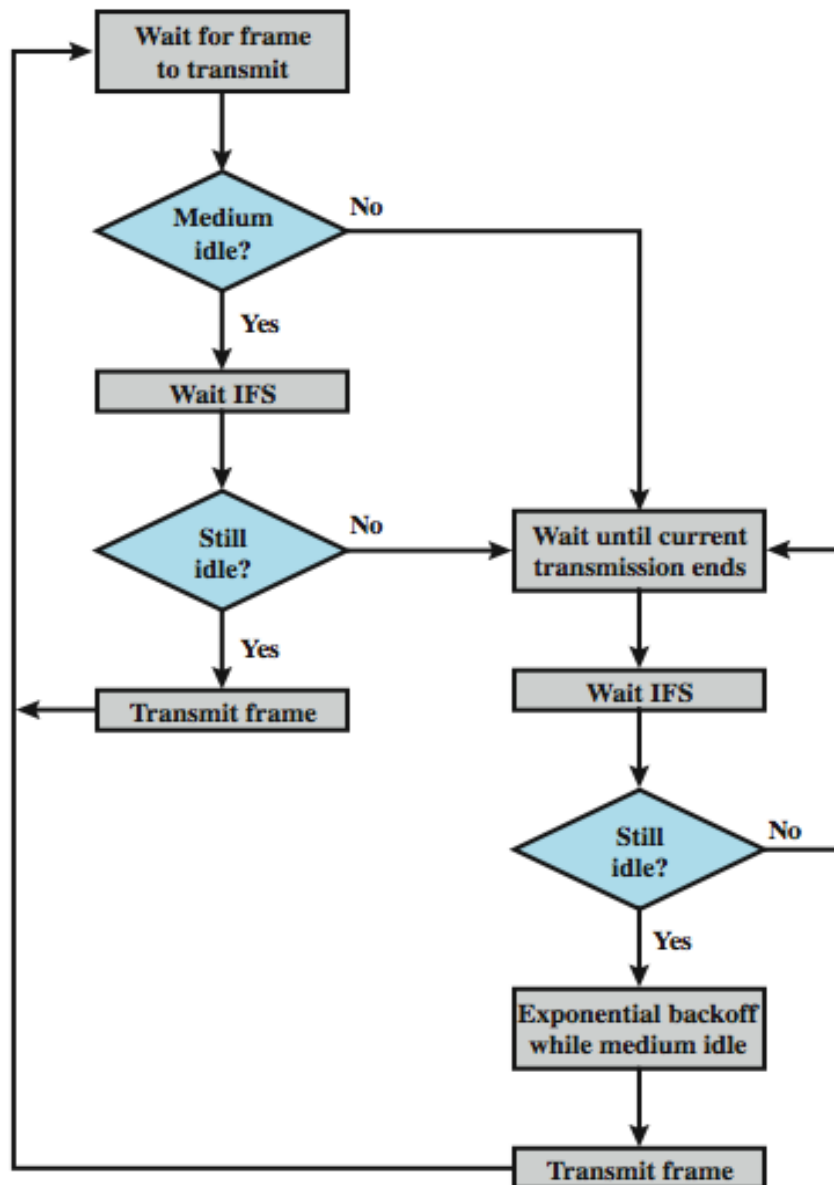
CSMA/CD



CSMA/CA

IEEE 802.11 MAC Protocol

IFS : Inter Frame Spacing



CSMA/CA Description

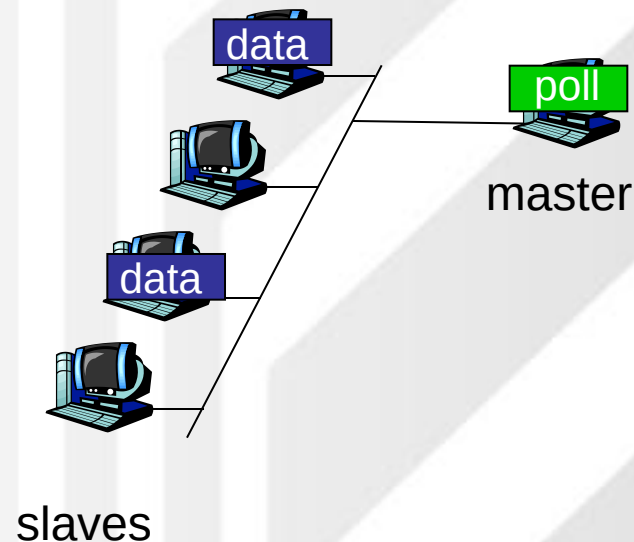
CSMA/CA rules:

1. if medium idle, (not send directly) wait for some IFS time, then sense the channel
 1. If busy – go to step 2
 2. If still idle – transmit a frame (but wait for contention window time)
2. if busy, wait until transmission end and then wait for some IFS time, then sense the channel again
 1. If busy – go to Step 2
 2. If idle – backoff/wait for some random time and transmit a frame

“Taking Turns” MAC protocols

Polling:

- master node
“invites” slave
nodes to transmit
in turn



“Taking Turns” MAC protocols

Token passing:

- control **token** passed from one node to next sequentially.





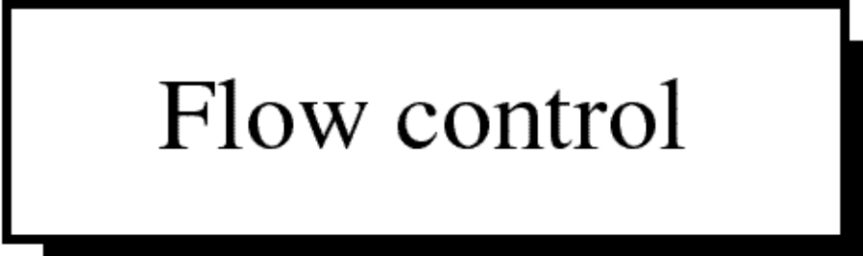
Summary of Access Control

- *channel partitioning*, by time, frequency or code
 - Time Division, Frequency Division
- *random access* (dynamic),
 - ALOHA, S-ALOHA, CSMA, CSMA/CD
 - carrier sensing: easy in some technologies (wire), hard in others (wireless)
 - CSMA/CD used in Ethernet
 - CSMA/CA used in 802.11
- *taking turns*
 - polling from central site, token passing
 - Bluetooth, FDDI, IBM Token Ring

Flow Control

- Procedures that tells the sender how much data to transmit.
- Each receiving device has a block of memory.
- Two methods
 - Stop-and-wait
 - Sliding window

Flow control



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graph TD; A[Flow control] --> B[Stop and wait]; A --> C[Sliding window];
```

Stop and wait

**Send one frame
at a time**

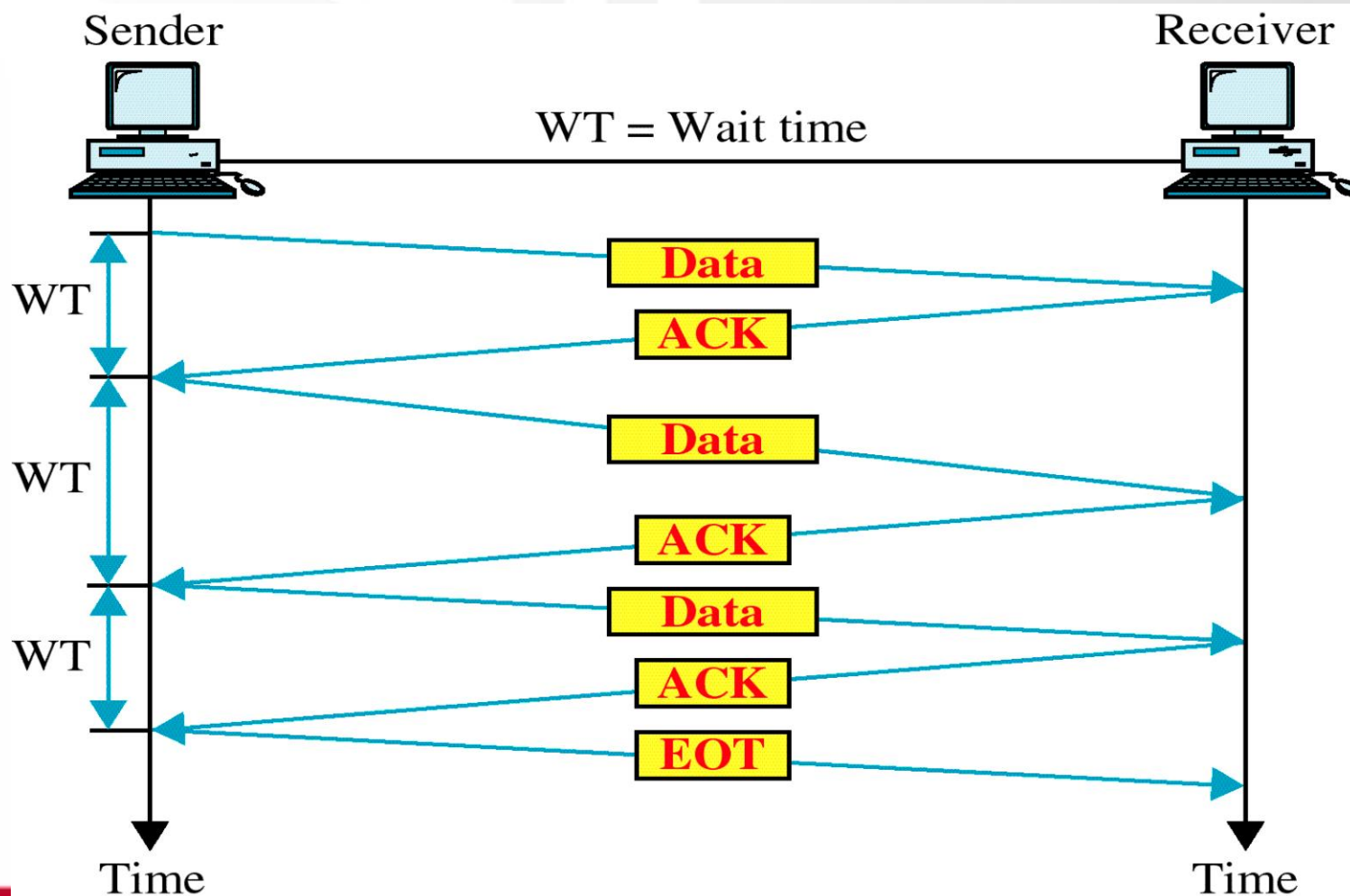
Sliding window

**Send several frames
at a time**

Stop and Wait

- source transmits frame
- destination receives frame and replies with acknowledgement (ACK)
- source waits for ACK before sending next
- destination can stop flow by not send ACK
- works well for a few large frames
- Stop and wait becomes inadequate if large block of data is split into small frames

Stop and Wait Link Utilization

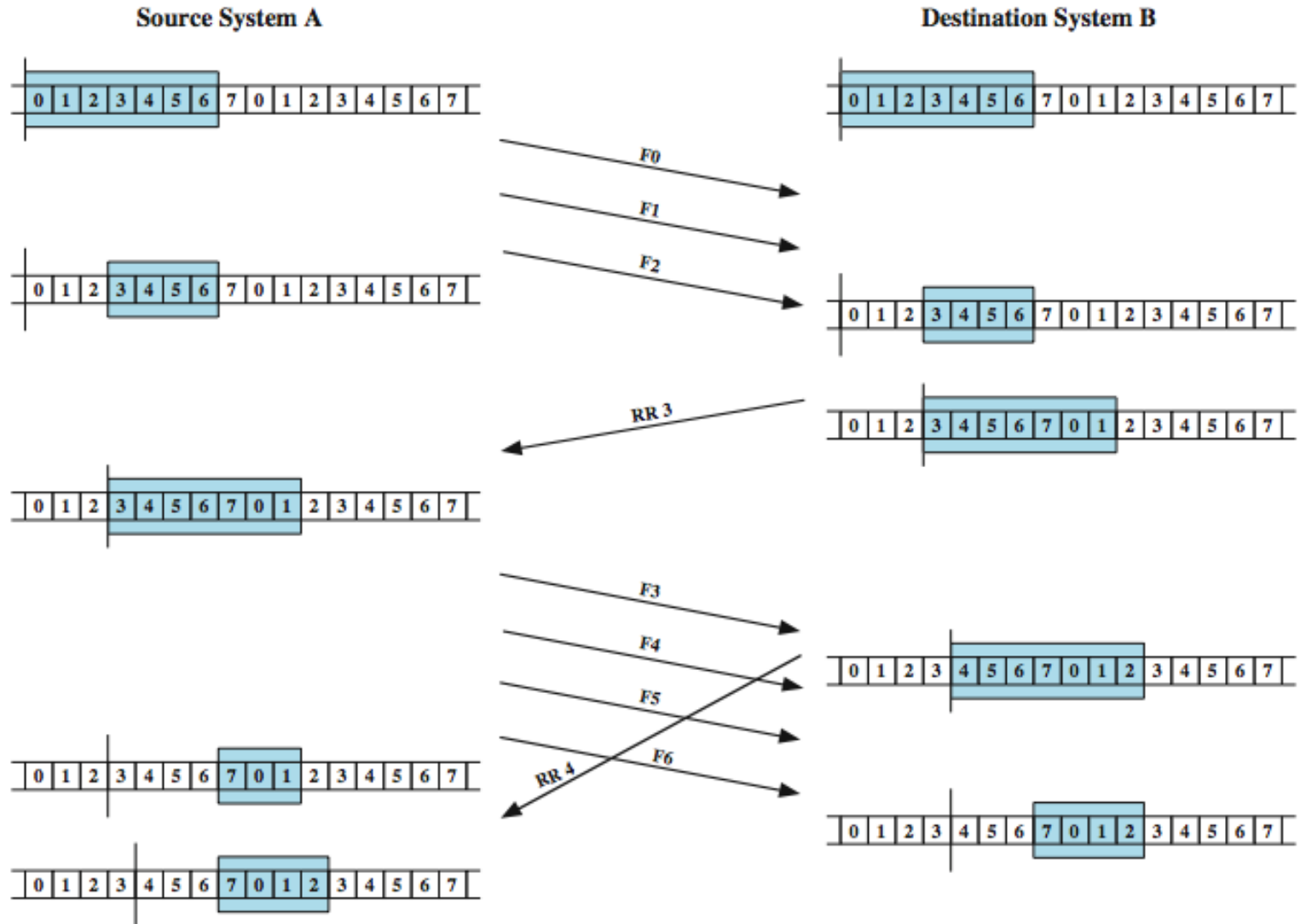




Sliding Windows Flow Control

- allows multiple numbered frames to be in transit
- receiver has buffer W long
- transmitter sends up to W frames without ACK
- ACK includes number of next frame expected
- sequence number is bounded by size of field (k)
 - frames are numbered modulo 2^k
 - giving max window size of up to $2^k - 1$
- receiver can ack frames without permitting further transmission (Receive Not Ready)
- must send a normal acknowledge to resume
- if have full-duplex link, can piggyback ACKs

Sliding Window Example





Error Control

- Refers to methods of error detection and correction due to
 - lost frames
 - damaged frames
- common techniques use:
 - error detection (CRC,checksum)
 - positive acknowledgment
 - retransmission after timeout
 - negative acknowledgement & retransmission

Automatic Repeat Request (ARQ)

- collective name for such error control mechanisms, including:
- stop and wait
- go back N
- selective repeat (selective retransmission)



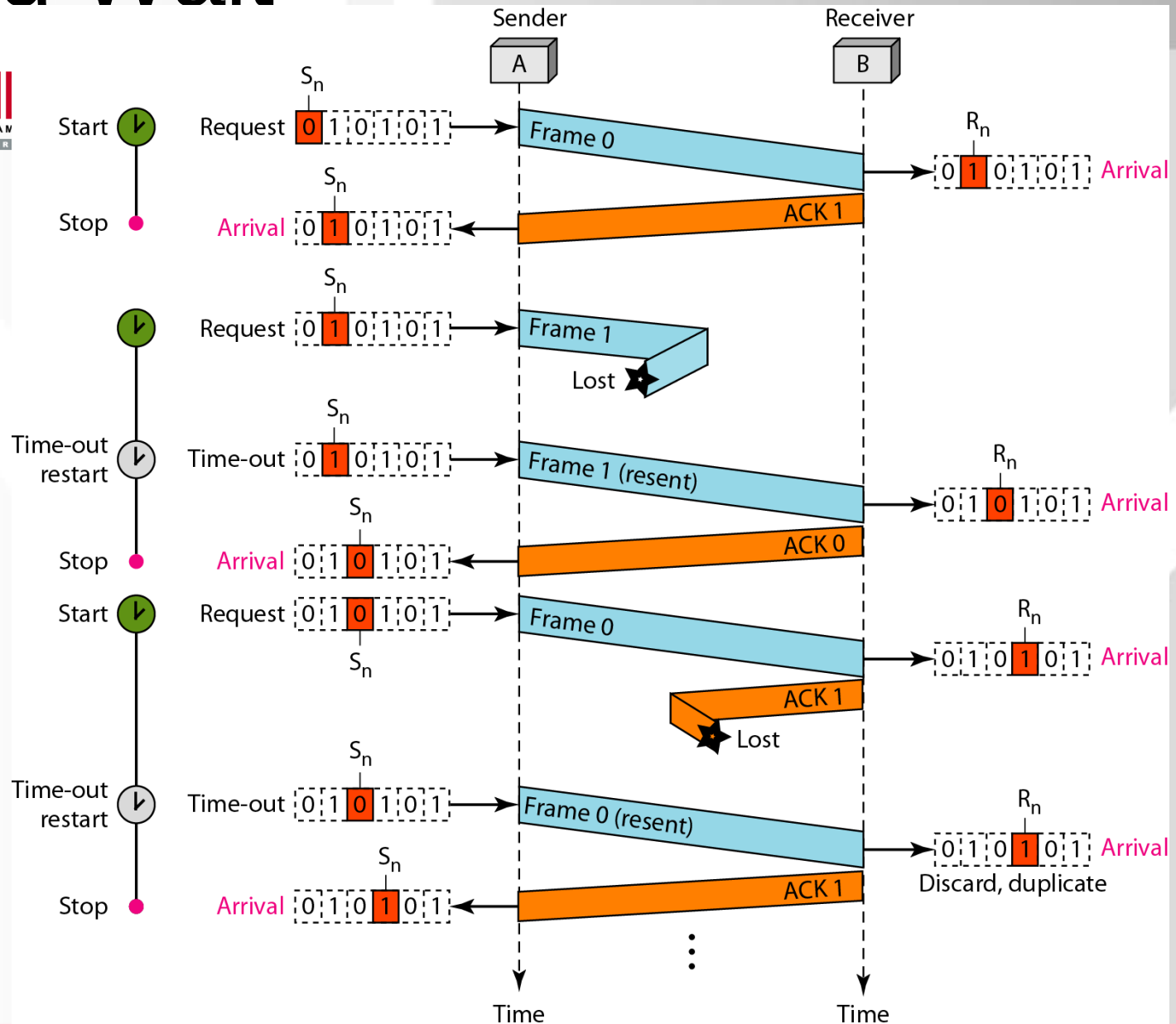
Stop and Wait

- source transmits single frame
- wait for ACK
- if received damaged frame, discard it
 - no ACK received
 - if no ACK within timeout, retransmit
- if ACK damaged, transmitter will not recognize it
 - transmitter will retransmit
 - receiver gets two copies of frame
 - use alternate numbering and ACK0 / ACK1

Stop and Wait



- see example with both types of errors
- pros and cons
 - simple
 - inefficient

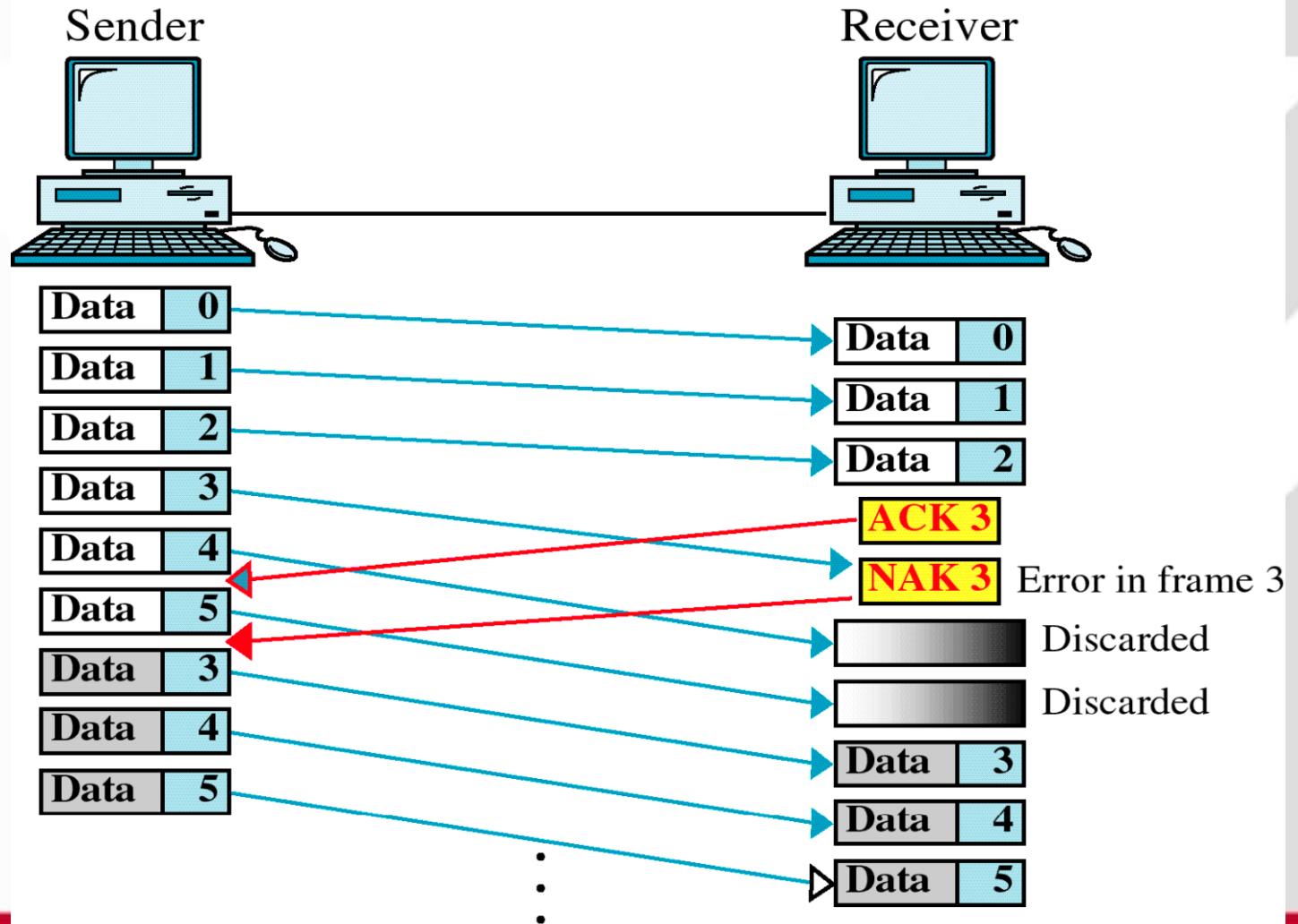


Go-Back-N

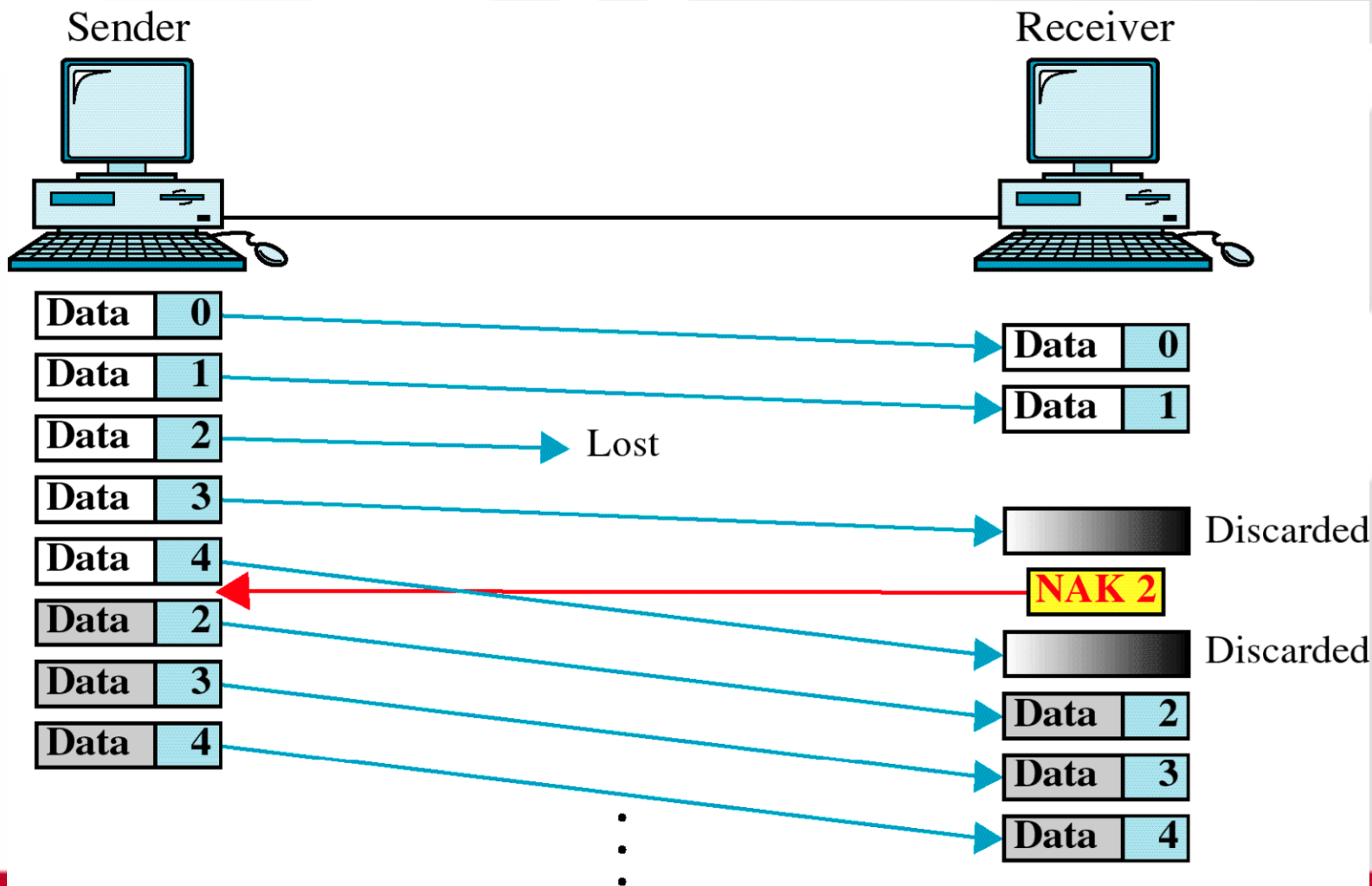
- based on sliding window
- if no error, ACK as usual
- use window to control number of outstanding frames
- if error, negative acknowledgment
 - discard that frame and all future frames until error frame received correctly
 - transmitter must go back and retransmit that frame and all subsequent frames



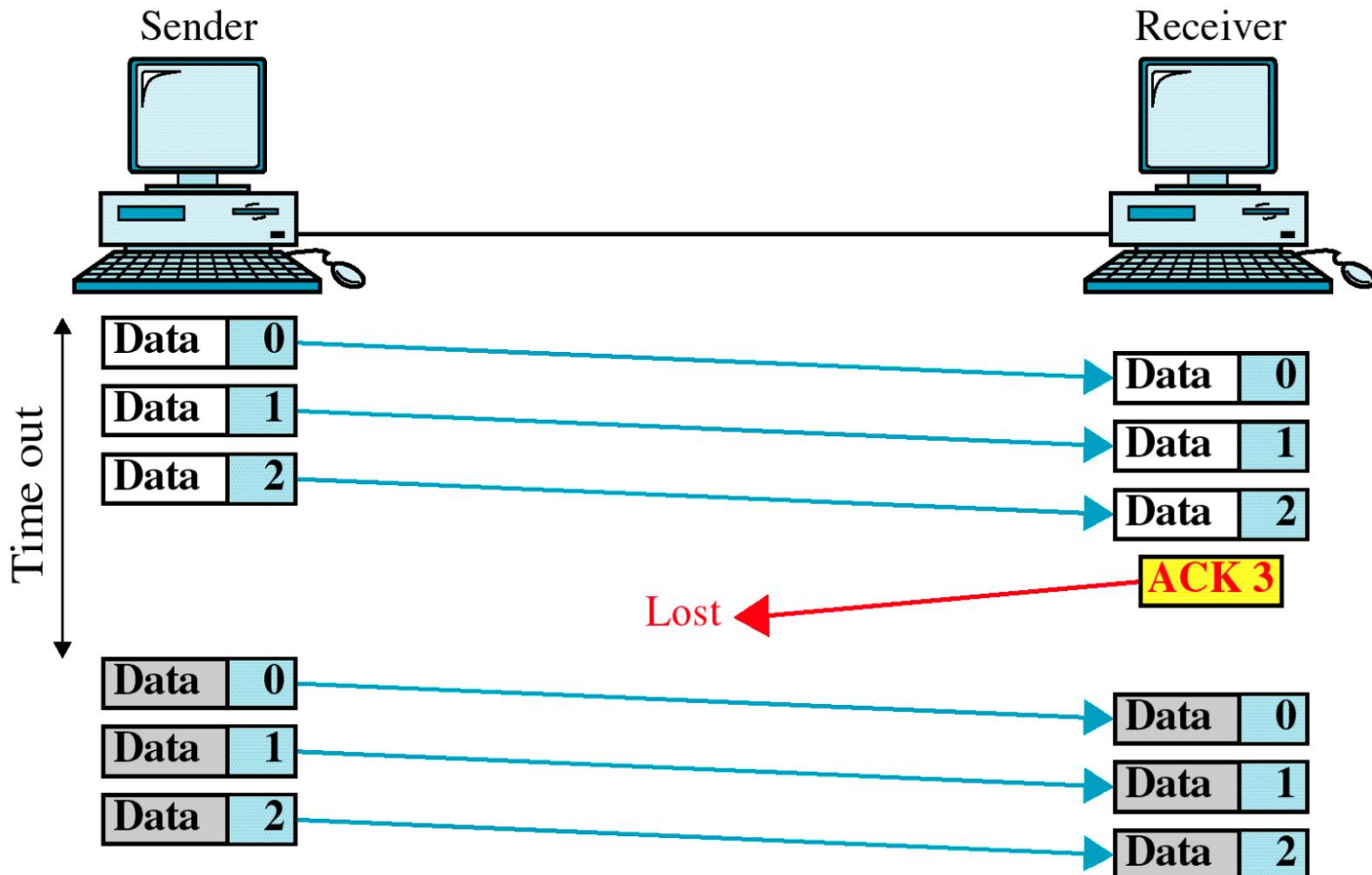
Damaged Frame



Lost Frame



Lost ACK



Selective Repeat

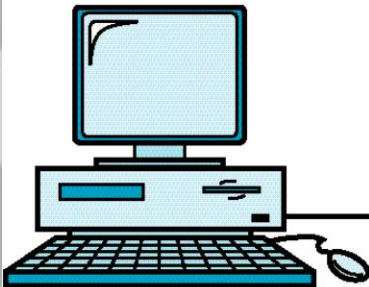
- only rejected frames are retransmitted
- subsequent frames are accepted by the receiver and buffered
- minimizes retransmission
- receiver must maintain large enough buffer
- more complex logic in transmitter
- hence less widely used
- useful for satellite links with long propagation delays



UPM

Selective Repeat

Sender



Receiver



Data 0

Data 1

Data 2

Data 3

Data 4

Data 5

Data 2

Data 0

Data 1

NAK 2

Data 3

Data 4

Data 5

Data 2

Resent